LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 9 Test: Light – Reflection and Refraction

Chapter / Topic: Light – Reflection and Refraction

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Use the New Cartesian Sign Convention. Show the formula, substitution and units in numericals. Draw ray diagrams neatly.

SECTION A

6 marks

Choose the correct option.

Q1. Which of the following can make a parallel beam of light when light from a point source is incident on it? **[1]**

(a) Convex mirror and concave lens
(b) Two plane mirrors at 90° to each other
(c) Concave mirror and concave lens
(d) Concave mirror and convex lens

Q2. The image formed by a concave mirror is virtual, erect and larger than the object. Where is the object placed? **[1]**

(a) Between F and C (b) At C
(c) Beyond C (d) Between the pole P and F

Q3. The focal length of a lens is +50 cm. Its power is: **[1]**

(a) −2 D (b) +0.5 D
(c) +2 D (d) +50 D

Q4. The refractive index of glass is 1.5. The speed of light in glass is (speed of light in vacuum = 3×10^8 m/s): **[1]**

(a) 1.5×10^8 m/s (b) 2×10^8 m/s
(c) 3×10^8 m/s (d) 4.5×10^8 m/s

Q5. The mirror used as a rear-view mirror in vehicles is: **[1]**

(a) convex (b) plane
(c) concave (d) any of these

Q6. **Assertion (A):** The magnification produced by a convex mirror is always positive and less than 1.
Reason (R): A convex mirror always forms a virtual, erect and diminished image. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Define the principal focus of a concave mirror. The radius of curvature of a spherical mirror is 20 cm. What is its focal length? **[2]**

OR

State the two laws of refraction of light.

Q8. A ray of light travelling in air enters water at an angle. Does it bend towards or away from the normal? Why? **[2]**

SECTION C

6 marks

Q9. An object 4 cm high is placed 15 cm in front of a concave mirror of focal length 10 cm. **[3]**

(i) Find the position of the image. **[2]**

(ii) Find the size and nature of the image. **[1]**

Q10. A concave lens has a focal length of 15 cm. At what distance from the lens should an object be placed so that the image is formed 10 cm from the lens? Find the magnification. **[3]**

OR

An object is placed 15 cm from a convex lens of focal length 10 cm. Find the position, nature and magnification of the image.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Draw a ray diagram to show the image formed by a convex lens when the object is placed between F_1 and $2F_1$. State the position and nature of the image. **[3]**

(ii) Define 1 dioptre. Find the power of a concave lens of focal length 2 m. **[2]**

OR

(a) Give two uses each of concave and convex mirrors. [2]
(b) An object 5 cm in length is held 25 cm away from a converging lens of focal length 10 cm. Find the position, size and nature of the image. [3]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. The refractive index of a medium tells us how much light slows down in it compared with a vacuum. Some values are: water 1.33, kerosene 1.44, crown glass 1.52 and diamond 2.42. **[4]**

(i) In which of these four media does light travel fastest? **[1]**

(ii) Which of these is optically the densest? **[1]**

(iii) Calculate the speed of light in diamond. (Speed of light in vacuum = 3×10^8 m/s.) **[2]**

~ All the best! ~